

Statistical Theory of Bimolecular Exchange Reactions*

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A new concise derivation of the statistical theory of exchange reactions is given. The results of the theory are compared with experiment for several reactions and for several types of information, and the reliability of such statistical calculations is assessed.

The area of gas-phase collision phenomena, both energy transfer and chemical kinetics, has become the subject of intensive experimental and theoretical investigation in recent years. Since the recent review articles¹⁻³ are both excellent and up to date, we shall concentrate, in this paper, only on the statistical (or phase space) theory of such phenomena. It is, however, the detailed experimental information about these processes, provided by various complementary experimental techniques, which has provided the impetus for further theoretical investigations.

The experimental information available about some reactions such as the alkali metal + halogen⁴⁻⁶ and the hydrogen + halogen exchange reactions⁷ is rapidly approaching the ultimate of providing cross sections for reaction as a function of velocity from a specified state (vibration-rotation) to a specified product state. In any case, it is quantitative theoretical understanding on this level which now seems to be the problem. If reasonable quantitative estimates of these cross sections could be made, all the information desired could be obtained by appropriate averages.

The theoretical approaches to this problem fall into three rather well-defined categories: quantum calculations,^{8,9} classical dynamical calculations,¹⁰⁻¹³ and approximate statistical calculations.¹⁴⁻¹⁷ Although quantum mechanics and the formal apparatus of scattering theory provides the most rigorous approach to the problem, they have not yet provided a large number of accurate results. There are two basic reasons for this: (a) the quantum dynamics of three-body (atomic size) systems are complex, often involving large numbers of states and strong interactions, and (b) the potential energy surfaces (presuming electronically adiabatic reactions) can only be estimated in a gross manner.

The most successful approach to date has been the second; following classical trajectories on various potential energy surfaces to determine the most essential features affecting the dynamics of chemical reactions. Since the pioneering work of Wall, Hiller, and Mazur,¹⁸ and Bunker,¹⁰ considerable information and confidence in the method have been produced. Although this approach has been successful, there are two limitations inherent in the method. First, specific quantum effects cannot be obtained. This is, almost certainly, a spurious criticism for most chemical reactions. The fact that many quantum states are normally involved and available almost guarantees that classical mechanics is a very good approximation. The second, more serious limitation is to electronically adiabatic potential energy surfaces.

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If more than one surface is present, a transition probability between surfaces must be assigned, normally zero or one. Although this has not been a problem in the reactions considered to date, there is a large class of chemical reactions which are not adiabatic electronically and which do produce electronically excited products.¹⁹

The statistical approach to chemical kinetics is based on the hope that it may be possible to bypass the complexities of quantum scattering theory and the uncertainties and labour of classical calculations (which consume non-trivial amounts of computer time) and yet to obtain quantitative results in good agreement with experiment. The statistical approach substitutes a hypothesis for much calculation, and the results must be subjected to comparison with experiment in order to test the validity of the hypothesis. The comparison must be made in such a way that at worst the hypothesis will be shown to be completely wrong, and at best it can be shown to be reasonable for a well-defined class of reactions.

There are several reasons for doing this. The most obvious is that it may provide a relatively simple method of obtaining reasonable answers, particularly for complex situations in which the other methods cannot be applied. In addition, the hypothesis is complementary to the other approaches. If, for a particular reaction, or class of reactions, the statistical approach seems to yield valid results, it implies that the system can in no way be considered as weakly coupled. Thus the quantum approach and the classical trajectory approach probably will be even more difficult to carry through successfully for these reactions.

There have now been enough calculations using the statistical theory and comparisons with experiment to provide a somewhat ambiguous answer as to its validity. In the following sections we shall try to provide a comprehensive outline of the foundations, applications and results of the statistical approach, summarizing, at this date, its validity and applicability.

FOUNDATIONS

All statistical theories of dynamical events, from nuclear reactions to chemical kinetics, involve two distinct types of approximations. The first is a definition, normally in phase space, of the limits of a strong coupling complex. The second is a statistical prescription, again usually involving phase space, for partitioning the products of the collision among the various possible products (or states).

Statistical models differ from thermodynamic models of rate constants (such as "absolute" reaction rate theory²⁰) in several important respects. First, the statistical theory is set up to describe the probabilities of possible outcomes of single well-defined collisions rather than the averages over equilibrium ensembles of these probabilities. Thus, in a statistical theory one may ask for the cross section at a particular initial velocity for the transition from initial state i to final state j . This question has no meaning in a thermodynamic theory of kinetics. Secondly, in a statistical theory one is forced to make assumptions only on the limits of a strong coupling complex but not on the lifetime or the configuration involved. By contrast, in a thermodynamic treatment, the complex or activated state must formally be treated as a well-defined "molecule", having a particular unobservable equilibrium configuration with known (but unobservable) vibration-rotation states. Thus the absolute reaction rate theory is parameterized to a much higher degree than is necessary in a statistical theory. This makes a meaningful comparison of the predictions of the statistical theory with experiment easier than that for the absolute reaction rate theory.

A statistical theory of dynamical events was first proposed by Fermi²¹ to describe multiple particle production in high energy nuclear collisions. This theory has been

the basis for much subsequent work in nuclear physics. It is, however, different in detail from the statistical theory of chemical kinetics described below.

In 1958 Keck,¹⁴ while investigating the kinetics of recombination, gave a clear, correct description of the elements of a statistical theory of chemical reactions in terms of the fluxes in phase space. This theory is equivalent to the corrected version of the phase space (or statistical) theory presented by the author and co-workers in 1965,²²⁻²⁶ although the language is different.

The statistical theory of kinetics may be constructed in several ways: classically, from the S -matrix, etc. We shall give here a brief derivation of the quantum mechanical version, emphasizing only the roles of symmetry and conservation requirements about which there has been some discussion.²⁷ More detailed derivations can be found in the literature.^{23, 28}

We shall consider the problem of a bimolecular exchange reaction involving only three atoms. We denote by α, β, γ , etc., the possible chemical species (channels) which can be formed in this reaction (α denotes $AB+C$, for instance). We denote the quantum numbers of the internal states of the diatomic molecule in channel α by s_α , the total rotational and orbital angular momenta by L_α and L respectively, and the relative translational energy in this channel by E_α .

To proceed to the essence of the statistical theory we now need to make one definition and one assumption. First, a "strong coupling complex" is defined by a single property:

(a) the mode of decomposition of a "strong coupling complex" is uncorrelated with the mode of formation except through conservation laws and detailed balancing. The conservation laws are conservation of energy, total angular momentum and its projection on one axis, and linear momentum. If so desired, one could also include electron spin angular momentum as being (approximately) separately conserved. Whether or not such complexes exist, even as an average over unobserved quantities is open to question (and experimental verification). The lack of correlation expressed in (a) is, however, a fundamental assumption of a statistical, as opposed to deterministic, theory. In the work of Keck and the author (although not in that of Eu and Ross²⁷), one further assumption is made which, although not absolutely necessary, seems reasonable and greatly simplifies (indeed permits) calculations. It can be stated as follows:

(b) the probability of formation of a "strong coupling complex" from every channel α is either unity or zero, depending upon the values of E_α, L, L_α and s_α . Thus

$$P_{\text{form}} = \begin{cases} 1 & L \leq L_{\max}(E_\alpha, s_\alpha, L_\alpha) \\ 0 & L > L_{\max}(E_\alpha, s_\alpha, L_\alpha) \end{cases}$$

The total cross section for formation of a "strong coupling complex" from the collision specified by $\alpha, E_\alpha, L_\alpha, s_\alpha$, is then

$$\begin{aligned} \sigma_T(\alpha, E_\alpha, s_\alpha, L_\alpha) &= \frac{\pi \hbar^2}{2\mu_\alpha E_\alpha} \sum_{L=0}^{L_{\max}} (2L+1) \\ &= \frac{\pi \hbar^2}{2\mu_\alpha E_\alpha} (L_{\max} + 1)^2 \end{aligned} \quad (1)$$

The classical analogue of this equation is obtained by equating, for large $L_{\max}$, the classical and quantum mechanical expressions for orbital angular momentum ($\hbar L = \mu v b$),

$$\sigma_{T(\text{classical})}(\alpha, E_\alpha, L_\alpha, s_\alpha) = \pi b_{\max}^2(\alpha, E_\alpha, L_\alpha, s_\alpha), \quad (2)$$

where b is the impact parameter.

Assumption (b) is certainly an approximation. Quantum mechanics, shows clearly that except for singular potentials, probabilities are “never” one or zero, i.e., there is always some penetration of a finite potential barrier into a non-classical region and a finite probability of reflection of a particle with enough energy to pass over a barrier. However, for sufficiently massive particles, such as atoms, approximations such as (b) are numerically very accurate except for $L \sim L_{\max}$, i.e., except for nearly orbiting collisions which constitute a minor fraction of the total number. Thus (b) is, in essence, a “classical” approximation. It implies that the density of states of the “complex” is so high that *all* collisions satisfying (b) can be accommodated.*

We shall now indicate briefly how statements (a) and (b) are sufficient to determine the statistical theory of chemical kinetics, specifically for bimolecular exchange reactions. The argument would be trivial in a two-dimensional world in which only one orientation of every angular momentum were allowed. In real space of three dimensions the coupling of L and L_α to form J make the algebra more complicated but do not change the basis of the derivation. We shall show that statements (a) and (b), together with the principle of detailed balancing, uniquely determine the probabilities of decomposition of the complex into any final state.

On this basis, detailed balancing requires that

$$(2L_\alpha + 1)\mu_\alpha E_\alpha \sigma(\{\alpha\} \rightarrow \{\beta\}) = (2L_\beta + 1)\mu_\beta E_\beta \sigma(\{\beta\} \rightarrow \{\alpha\}), \quad (3)$$

where μ_α is the reduced mass of the atom-diatomic molecule in channel α , L_α the rotational angular momentum, and $\{\alpha\}$ stands for the entire set of quantum numbers E_α , s_α , L_α . Because of the definition (a), we may write the cross sections as products of a cross section for formation of the complex and the probabilities of decomposition of the complex. Thus, for a given total energy,

$$\sigma(\{\alpha\} \rightarrow \{\beta\}) = \sum_J \sigma(\{\alpha\} \rightarrow J) P(J \rightarrow \{\beta\}), \quad (4)$$

where J is the total angular momentum of the complex.

The cross section for formation of a complex of specified J is, from eqn. (1),

$$\sigma(\{\alpha\} \rightarrow J) = \frac{\pi \hbar^2}{2\mu_\alpha E_\alpha} \sum_{L(\{\alpha\}, J)} (2L + 1) \left\{ \frac{(2J + 1)}{(2L + 1)(2L_\alpha + 1)} \right\}, \quad (5)$$

where the last factor is the fraction of collisions with L , L_α which couple to yield a total angular momentum J .

The summation over L is restricted by the stronger of the two conditions (a) that $|L - L_\alpha| \leq J \leq |L + L_\alpha|$, and (b) that $L \leq L_{\max}(\alpha, E_\alpha, s_\alpha, L_\alpha)$.

Combining eqn. (5), (4) and (3), we have

$$\pi \hbar^2 \sum_J (2J + 1) \sum_{L(\{\alpha\}, J)} P(J \rightarrow \{\beta\}) = \pi \hbar^2 \sum_J (2J + 1) \sum_{L(\{\beta\}, J)} P(J \rightarrow \{\alpha\}). \quad (6)$$

Since $P(J \rightarrow \{\beta\})$ is, by assumption (a), independent of $\{\alpha\}$, we have

$$\pi \hbar^2 \sum_J (2J + 1) P(J \rightarrow \{\beta\}) \sum_{L(\{\alpha\}, J)} 1 = \pi \hbar^2 \sum_J (2J + 1) P(J \rightarrow \{\alpha\}) \sum_{L(\{\beta\}, J)} 1. \quad (7)$$

The last sums are equal, by definition, to the numbers of states available in the specified channel which can form, or be formed from, a complex of specified energy and angular momentum:

$$\sum_{L(\{\alpha\}, J)} 1 \equiv n[\{\alpha\}, J]. \quad (8)$$

* This is not true for electron-atom collisions in which well-defined resonances have been observed. However, no such resonances have been observed in chemical reactions.

It may now be proved by induction, starting with $J = 0$, that the only reasonable solution of (7) is the obvious one:

$$\frac{P(J \rightarrow \{\beta\})}{n[\{\beta\}, J]} = \frac{P(J \rightarrow \{\alpha\})}{n[\{\alpha\}, J]} = \text{constant.} \quad (9)$$

The requirement that the sum of the probabilities of decomposition of any complex be unity provides the normalization and leads to the final result:

$$P(J \rightarrow \{\alpha\}) = \frac{n[\{\alpha\}, J]}{\sum_{K=\alpha, \beta, \gamma} n[\{K\}, J]} = \frac{n[\{\alpha\}, J]}{N_{\text{tot}}(J)}, \quad (10)$$

where the summation in the denominator runs over all states accessible from the specified complex. Thus N_{tot} is the total number of states available from a complex of given total energy and angular momentum. Thus the cross section for the overall process is

$$\begin{aligned} \sigma(\{\alpha\} \rightarrow \{\beta\}) &= \sum_J \sigma(\{\alpha\} \rightarrow J) \frac{n[\{\beta\}, J]}{N_{\text{tot}}(J)} \\ &= \frac{\pi \hbar^2}{2\mu_\alpha E_\alpha (2L_\alpha + 1)} \sum_J (2J + 1) \frac{n[\{\alpha\}, J] n[\{\beta\}, J]}{N_{\text{tot}}(J)}. \end{aligned} \quad (11)$$

In the derivation the relative density of translational states enters only through the energy dependence of the cross sections and the microscopic reversibility relation. This is because we have dealt with probabilities rather than fluxes. However, the two results would be equivalent. Let P_r be the *radial* relative momentum (the only free variable since the angular momenta are specified). The flux into each state per unit energy range will be the velocity times the probability times the density of radial translational states per unit energy. Thus we have

$$\frac{d}{dE_\alpha} \text{flux}(J \rightarrow \{\alpha\}) = v_r(\{\alpha\}) P(J \rightarrow \{\alpha\}) \frac{dp_r(\{\alpha\})}{dE_\alpha} = P(J \rightarrow \{\alpha\}). \quad (12)$$

The above theory could be modified by changing one or both of the statements (a) and (b) to allow, for instance, for preferential coupling of certain initial and final states. This could most easily be done in the S matrix formulation of the theory.²⁸ However, unless such modifications were made with a firm theoretical basis their value would be open to question since it would correspond to *ad hoc* parameterization. The purpose of such a theory is not to fit every experiment but to give a general model for which *a priori* calculations can be made and used for comparison and (hopefully) reasonably accurate prediction. Large scale tampering with the model serves neither purpose.

THE "STRONG COUPLING COMPLEX"

In the preceding section we used only the formal properties of the "strong coupling complex" in the general derivation of the statistical theory. In this section we shall describe briefly how the limits of the complex, which are needed before any computation can be done, can be chosen. Since the choice of the limits to be used in statement (b) completely determines all the cross sections, this is the only point at which the physics of the process under consideration can be inserted. Again one wants to choose the simplest limits which seem physically reasonable.

For reactions without activation energy we have reasoned as follows, using the classical impact parameter for simplicity. There should be a distance $r_0(\alpha)$ for each

channel within which the atom and molecule are strongly coupled. Since there is no activation energy, r_0 should be approximately independent of the internal and translational energy and the rotational angular momentum. Thus the limit of the strong coupling complex required in (b) is the maximum impact parameter, $b_{\max}(E_\alpha)$ for which the atom will approach within $r_0(\alpha)$ of the molecule. For low energies with long-range attractive forces between the atom and molecule and with r_0 about equal to the range of the repulsive (if any) forces, this leads to

$$b_{\max}(E_\alpha) = C_\alpha/(E_\alpha)^\delta, \quad (13)$$

where C_α and δ are determined by the form of the attractive potential (in each channel).²³ This form seems appropriate for ion-molecule reactions and for some neutral reactions such as $\text{K} + \text{Br}_2$, $\text{Cl} + \text{Na}_2$, and $\text{Hg}^*(^3P_0) + \text{H}_2$. At higher energies, however, $b_{\max}$ given above may be less than r_0 . This should not be allowed to happen and the correct limits for unit probability of formation of the strong coupling are

$$b_{\max}(E_\alpha) = \text{largest of } \begin{cases} C_\alpha/(E_\alpha)^\delta \\ r_0(\alpha) \end{cases} \quad (14)$$

Several calculations using eqn. (13) have been carried out for ion-molecule and neutral reactions. Some of the results will be given in the next section.

For reactions with activation energy, the limits for formation of the strong coupling complex must depend on translational energy and probably internal energy as well.²⁶ Only a few calculations have been made for these reactions and the definitions used were:

$$b_{\max}(E_\alpha) = \begin{cases} r_0(\alpha)[(E_\alpha - E_{\text{eff}}(\alpha))/E_\alpha]^{1/2} & E_{\text{eff}}(\alpha) \geq 0 \\ r_0(\alpha) & E_{\text{eff}}(\alpha) < 0 \end{cases} \quad (15)$$

Here the effective activation energy $E_{\text{eff}}(\alpha)$ was defined as the minimum translational energy necessary to have reaction for the molecule in a given internal state. The specific dependence on internal energy of the molecule was linear:²⁶

$$E_{\text{eff}}(\alpha) = E_A^0(\alpha) - \lambda_v E_{\text{vib}} - \lambda_{\text{rot}} E_{\text{rot}},$$

where E_A^0 is the activation energy from the ground state of the reactants, E_{vib} and E_{rot} are the vibrational and rotational energies of the reactant molecule and the λ are positive constants between 0 and 1. Although the parameterization in this case is higher than for reactions without activation energy, this form, which follows from simple collision theory of kinetics, seems as simple as is physically reasonable.

Before going on to results, a physical description of what a "strong coupling complex" might be seems in order. The mathematical properties are clear: once a system enters such a complex it "forgets" where it came from. In fact this does not happen. In practice, however, there are certain quantities associated with a collision which one cannot measure such as the z component of the orbital angular momentum, the magnitude of the orbital angular momentum, the exact translational energy, etc. If the coupling between all open channels is strong, the one might expect that cross sections, averaged over the unobserved quantities, to behave statistically. This would seem to be particularly likely if the complex lasts a reasonable length of time classically, i.e., for periods larger than the rotational period. Alternatively, if the potential gradients are large and not smooth, rather random scattering could occur. Thus the strong coupling approximation is essentially the opposite of an adiabatic approximation in which the system never changes state. We shall see in the next section that the adequacy of the theory depends, to a large extent, on the type of information desired. For highly averaged quantities the statistical theory works well but for

more detailed information the deviations from the statistical predictions become more apparent.

Finally the limits in statement (b) are not the actual limits of the strong coupling region itself. They are the limits, in terms of the asymptotic states of the system, within which a strong coupling complex will, at some time and in some place, be formed with unit probability. Thus they define asymptotic trajectories which, if followed, will lead through a strong coupling region. No information is required about the time spent in this region—only that it has the property defined in (a).

RESULTS

There has been a number of specific applications of the statistical theory to various reactive and inelastic processes.^{29, 30} We shall present here only four examples in any detail. These examples are chosen to illustrate the types of systems to which the theory can be applied, and the types of information which can be obtained and compared with experiment.

The four cases are $\text{H} + \text{Cl}_2$, $\text{K} + \text{HBr}$, $\text{He} + \text{H}_2^+$, and $\text{Hg}^* + \text{CO}$. These illustrate the application to distinct types of reactions: exchange reactions with activation energy, exchange reactions with negligible activation energy, ion-molecule reactions,

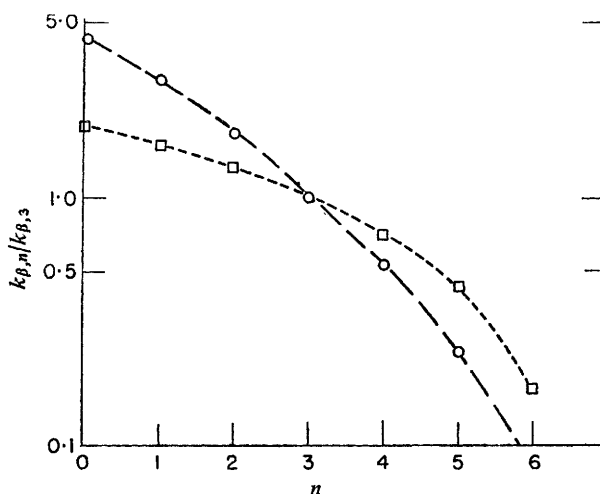


FIG. 1.—Relative reaction rates k_n/k_3 into vibrational states n for the reaction $\text{H} + \text{Cl}_2 \rightarrow \text{HCl}(n) + \text{Cl}$. -- □ -- calculated, $\lambda_v = \lambda_r = 1.0$; -- ○ -- calc., $\lambda_v = \lambda_r = 0.97$. (The experimental values for $n < 2$ are lower than for $n = 3$ (ref. (34)).

and excited atom-molecule reactions. These cases were all studied by the author and co-workers²²⁻²⁶ in an attempt to define the limits of applicability of the theory by comparison with experiment.

A. $\text{H} + \text{Cl}_2$ ²⁶

The luminescent studies of this reaction by Polanyi³¹⁻³³ and co-workers have provided unparalleled information about the internal excitation of the products. The reaction has an activation energy (of about 2.0 kcal/mole) and thus the definition of the complex in eqn. (15) was used in the statistical theory. The calculation was done with only the vibrational degree of freedom quantized, the angular momenta being treated classically. Since the experimental information was for thermal systems, all quantities were averaged over the appropriate Maxwell-Boltzmann distribution. The only significant parameters to enter the calculation are λ_v and λ_r , which measure

the fractional contribution of internal energy in reducing the effective activation energy, and the r_0 for each channel—the radii of the complex itself. The values of r_0 apparently have little effect in the statistical calculation except the gross cross sections. However, the values of λ affect the internal energy distributions considerably. The internal calculated energy distributions of the product are plotted in fig. 1 and 2.

These results illustrate one apparently general result. With complexes defined only by a single inequality on the impact parameter as a function of the energy variables, the statistical theory *never predicts complete population inversions* of products. There

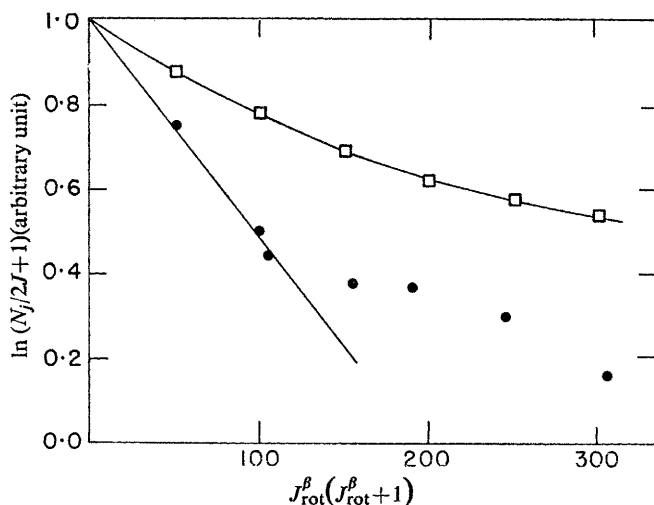


FIG. 2.—Rotational distributions of HCl. —□— calculated for HCl ($n=2$), $\lambda_v = \lambda_r = 1.0$, $E_\alpha = 0.1$ eV; ●, expt. (ref. (31)-(33)).

are always more states available from a lower vibrational state than from a high one and thus the populations of vibrational states of products fall monotonically with vibrational quantum number. This result has appeared in *every* statistical calculation to date. Since there is a large class of reactions proceeding with and without activation energy for which the best indications are that vibrational population inversions are produced, we find that the statistical theory is unreliable with respect to this type of information.

The rotational distribution given by the statistical theory cannot be directly compared with experimental results since no firm experimental quantities are known.³⁴ It seems, however, to be a reasonable distribution since high rotational excitation, after some relaxation, is still observed.³³ The overall rate constants as a function of temperature were computed. The result, from the statistical theory, is

$$k(T) = 2.1 \times 10^{-9} \exp(-2200/RT) \text{ cm}^3 \text{ sec}^{-1} \text{ molecule}^{-1}.$$

This seems in line with rates of similar reactions, but no good absolute values are available from experiment. One interesting point is that the value of 2.2 kcal/mole appearing in the exponential factor of the rate coefficient is 10 % larger than the activation energy E_A° assumed in the definition of the complex. This effect was also found in the studies of Karplus, Porter, and Sharma.¹³

B. $\text{K} + \text{HBr}$ ^{25, 26}

This reaction, studied experimentally by molecular beam methods,³⁵⁻³⁷ proceeds with an activation energy of less than 0.4 kcal/mole. Considerable information has

been deduced from the experiments about the details of the reaction: probability of reaction against impact parameter, rotational, vibrational, and total energy distributions of the products, etc. In this reaction, the value of $E_A^\circ = 0.15$ kcal/mole (best experimental estimate) was used and $\lambda_v = 0.8$ and $\lambda_r = 0.4$ were chosen to give

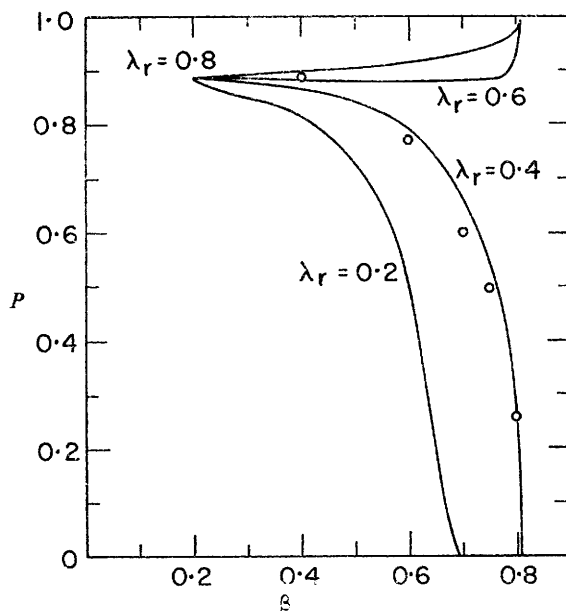


FIG. 3.—Probability of reaction against reduced impact parameter for K+HBr. $\lambda_v = 0.8$. ○, experimentally deduced points, ref. (35). $E_x = 0.08$ eV.

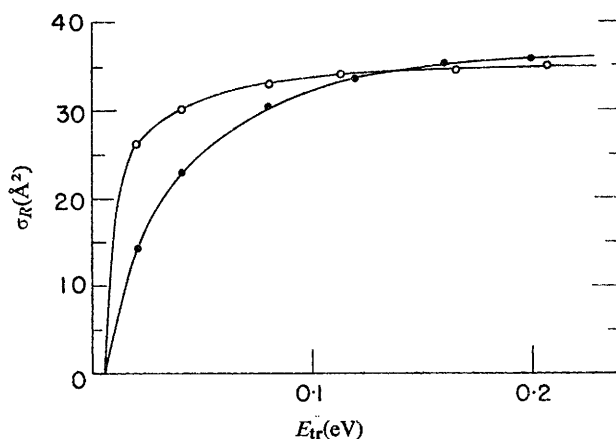


FIG. 4.—Reaction cross section for K+HBr against translational energy, E_{tr}° (eV). $\lambda_v = 0.8$, $\lambda_r = 0.4$. —○— experimentally deduced, ref. (35). —●—, calc.

the best agreement with the "experimental" probability of reaction against impact parameter. The value of $r^0 = 3.8$ Å for the product channel was taken from Suplinskas.³⁰ The aim of the calculation was to see how much of the experimental data could be reproduced by the statistical theory with one set of 5 parameters (λ_v , λ_r , E_A° , $e_{(a)}^\circ$, $r_{(\beta)}^\circ$).

In fig. 3-7 we plot the comparisons with experiment. It can be seen that for the probability of reaction against impact parameter, the reaction cross section against translational energy, and the rotational energy distribution, the agreement between

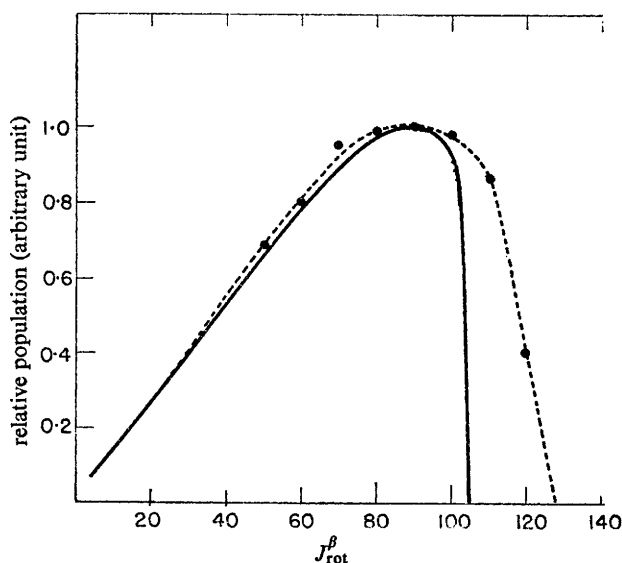


FIG. 5.—Rotational distribution of KBr produced. Curves normalized to the same maximum. --●-- experimentally deduced, ref. (12); — calc., $\lambda_v = 0.8$, $\lambda_r = 0.4$.

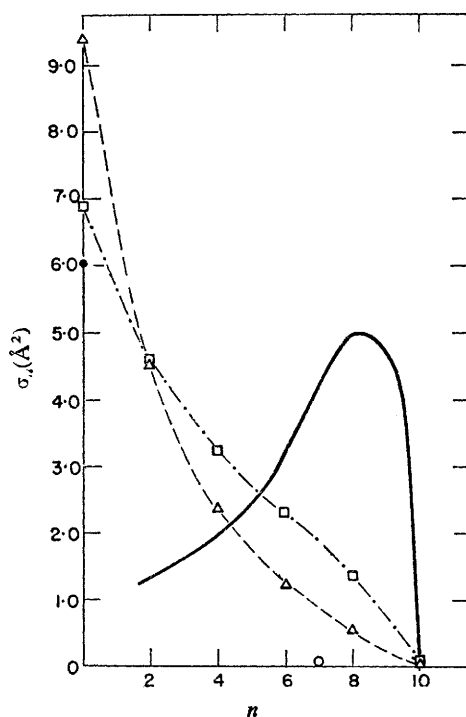


FIG. 6.—Vibrational distribution of KBr produced. —△— calc., $\lambda_v = 0.8$, $\lambda_r = 0.4$; —□— calc., $\lambda_v = 0.9$, $\lambda_r = 0.4$; — “Typical” Monte Carlo result, ref. (10).

the statistical calculation and the experimental estimates are good. We again find, however, that the vibrational distribution of the product molecule is probably not given correctly by the statistical theory, and this leads to a shift of the total internal energy distribution of the product to lower energy. This is sensitive to the assumed value of λ_v and a value of $\lambda_v = 0.9$ improves the agreement considerably, but does not reproduce the curve of the probability of reaction against impact parameter quite as well. Since the only quantities varied were λ_v and λ_r , we can say that for this reaction, with E_A° and the r° given, the values $\lambda_v = 0.8$ or 0.9 and $\lambda_r = 0.4$ reproduce well all the experimental information except, probably, the vibrational energy distribution.

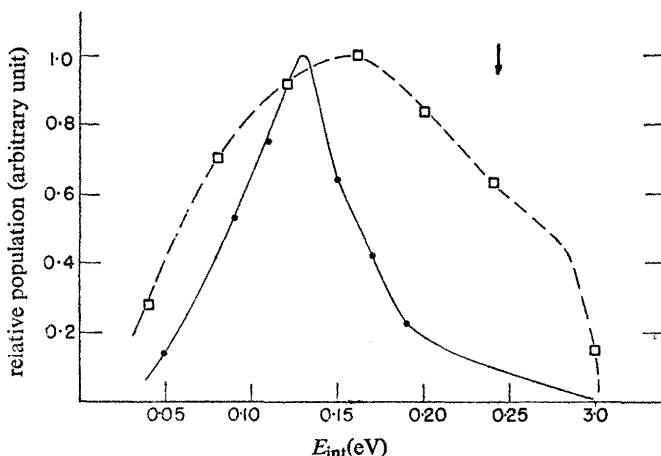


FIG. 7.—Total internal energy distribution of KBr produced. —□—□ calc., $\lambda_v = 0.9$, $\lambda_r = 0.4$; —●—●— calc., $\lambda_v = 0.8$, $\lambda_r = 0.4$. Curves normalized to maximum = 1 $E_a = 0.08$ eV. Arrow indicates experimental maximum (ref. (36)).

C. $\text{Hg}^* + \text{CO}$ ²⁶

The internal (vibrational) energy distribution for the CO produced in the quenching reaction of $\text{Hg}(^3P_0)$ has been studied by Karl, Kruse, and Polanyi.³⁹ The statistical theory, with a value of $\lambda_v = 0.98$ and an activation energy of about zero (0.005 eV) gives moderately good agreement with the experimental results as shown in fig. 8. That this model is not unique in giving reasonable results has been shown by Karl *et al.*, who used a model of impulsive energy release in a Hg^*CO complex. Additional credibility is lent to the statistical calculation for this type of reaction by the recent work of K. Yang and co-workers³⁰ on the $\text{Hg}^* + \text{H}_2$ reaction. They used the statistical theory to predict accurately the quantum yield of H atoms. Thus we may hope that excited atom-molecule interactions approximate the strong coupling complex with some degree of accuracy.

D. $\text{He} + \text{H}_2^+$ ²⁴

As one of the simplest ion-molecule reactions, this reaction, $\text{He} + \text{H}_2^+ \rightarrow \text{HeH}^+ + \text{H}$, has been studied in detail experimentally,⁴⁰⁻⁴³ although no information concerning energy distributions of products is available. For this reaction the limiting impact parameter for formation of the complex was taken from the Langevin⁴⁴ theory, i.e., it is determined by the requirement that the particles be able to pass *over* the angular momentum barrier. Since this is determined only by the polarizability of

the neutral species, there are *no adjustable parameters*. The initial vibrational distribution of the H_2^+ was taken from the calculations of Wacks.⁴⁵ It is critical since the reaction is endothermic from the ground state by 0.92 eV. Thus at low translational energies, only vibrationally excited H_2^+ ions can react.

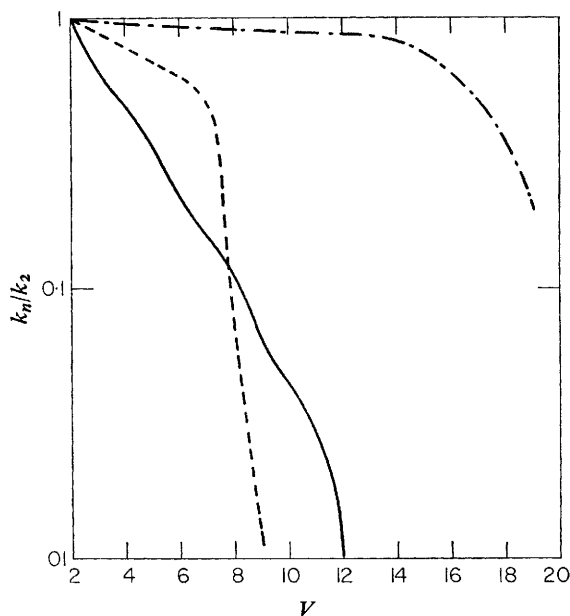


FIG. 8.—Vibrational energy distribution of CO from the reaction $\text{Hg}(^3P_0) + \text{CO} \rightarrow \text{Hg} + \text{CO}^*$. — calc., $\lambda_v = 0.98$. ---- calc., $\lambda_v = 1.0$. - · -, expt., ref. (39).

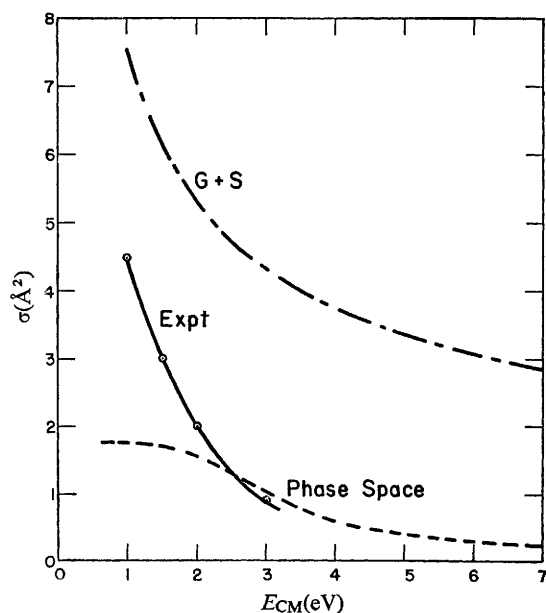


FIG. 9.—Cross section for $\text{He} + \text{H}_2^+ \rightarrow \text{HeH}^+ + \text{H}$ against E_{α} . Initial vibrational distribution ref. (45). --- calc., - · - total "strong coupling" cross section, —●— expt. ref. (41).

The cross sections for reactions from each vibrational state of H_2^+ were evaluated, and the overall cross section was obtained from the sum, weighted by the fractional population of each state. The results are compared with two sets of experimental data in fig. 9 and 10. In fig. 10, the calculated results were appropriately averaged

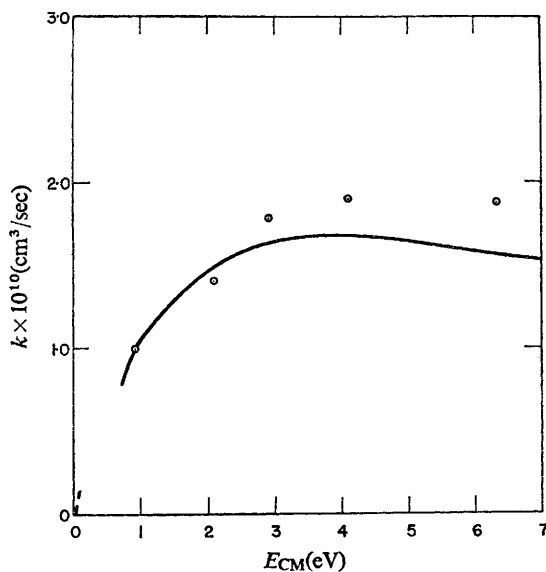


FIG. 10.— $\text{He} + \text{H}_2^+ \rightarrow \text{HeH}^+ + \text{H}$. Average rate coefficient against average translational energy (in mass spectrometer). — calc., $\bigcirc$ expt., ref. (42).

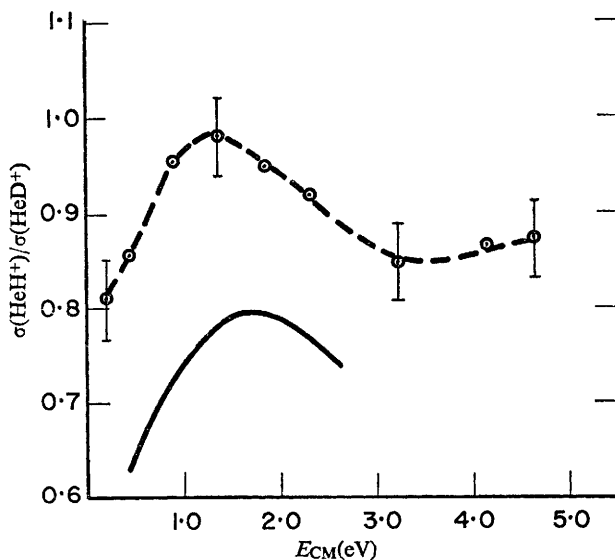
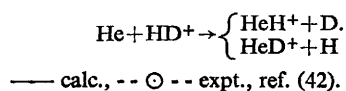


FIG. 11.—Average isotope ratio against average translational energy



to conform to the experimental velocity distribution in a conventional mass spectrometer. Of interest is the fact that the calculated cross sections fall off at low energy because of the purely statistical factor that more inelastic than reactive channels are open there. Thus the correct result is given without assuming that a minimum translational energy (activation energy) is necessary.⁴⁶ The isotope ratios for products of the $\text{He} + \text{HD}^+$ reaction were also calculated and compared with experiment. Fig. 11 shows that the qualitative dependence of this ratio on energy is reproduced well but the absolute value of the ratio is in error by approximately, 20%.

In general, the application of the statistical theory to low energy ion-molecule reactions has been productive. The most important effects seem to be predicted with reasonable accuracy (such as ratios of possible products²⁹ and velocity dependence of cross sections). It is likely that, because of the strong long range attractive forces involved, the ion-molecule reactions *at low energies* proceed through a complex which is strongly coupled. A good discussion of these processes is given by Futrell and Abramson.⁴⁰

SUMMARY AND DISCUSSION

The results presented in the previous section, while typical, represent only a small fraction of the cases studied by the statistical method. As in the examples above, the results, in comparison with experiment, vary from being qualitatively and quantitatively in error to being quantitatively correct. It is now possible to set certain limits on the applicability of the statistical theory. There are two factors which must be considered: first, the type of information desired, and second, the type of reaction involved.

In order to summarize the experience to date, we give, in table 1, the evaluation of the accuracy of the statistical theory for the specific types of reactions and information. The agreement is best for highly averaged quantities in ion-molecule reactions

TABLE 1.—RELATIVE ACCURACY OF STATISTICAL THEORY

information \ reaction	ion-molecule reactions ($E < 10$ eV)	excited molecule reactions	atom-molecule reactions no activation energy	atom-molecule reactions activation energy
thermal rate constants	good		good	fair
cross section against energy	good			good
isotope and product ratios	fair to good	good		
total internal energy of products		fair	fair	fair
rotational energy distributions			good	fair(?)
vibrational energy distributions		fair	poor	poor

which are likely to have the strongest coupling, and worst for detailed information about neutral reactions with activation energy, even though one parameter λ_v was varied to some extent.

There are three questions which it seems pertinent to attempt to answer (i) Are there any types of reactive systems for which the basic assumptions of the statistical theory, (a) and (b), are approximately valid in detail, i.e., on a state for state basis? (ii) Are there any types of information about specific types of reactions for which the statistical theory gives reliable quantitative results? (iii) When, if ever, should one use the statistical theory?

The answer to the first question is easy, but ambiguous. No reactive system has been shown to behave statistically in detail, but many have been shown to react in a

non-statistical manner. The only types of reactions which might be truly statistical are low energy ion-molecule reactions for which adequate experimental information to give a definitive answer is not available. Thus we may conclude that it is possible but not likely that any real systems react statistically in detail.

The answer to the second question is yes. Averaged information about ion-molecule reactions and excited atom-molecule reactions has usually been given accurately by the statistical theory.^{4,8} Quantities such as rate constants, cross sections against energy, and, perhaps most important, ratios of products can be computed reliably by the statistical theory. For reactions of neutral, unexcited species this holds true to a lesser degree.

The third question is, perhaps, the most important. There appear to be two reasonable uses of the statistical theory: as a model against which to compare experimental results, and as a predictive tool in the absence of much experimental information. The utility as a model was illustrated for the $\text{He} + \text{H}_2^+$ reaction. The fall-off in cross section for reaction as the translational energy is decreased is explained by the statistical theory as the statistical behaviour for an endothermic reaction proceeding from excited vibrational states. Although it cannot be asserted that this is the correct explanation, its simplicity and accordance with known facts of threshold behaviour give it added weight vis-a-vis an alternative explanation in terms of a kinetic energy threshold for ion-molecule reactions.^{4,6} In the opposite sense, the failure of the statistical theory to account for the vibrational energy distribution in the $\text{H} + \text{Cl}_2$ reaction means that a more deterministic model is both necessary and reasonable as the accurate approach.

Finally, the author believes that the statistical theory can be used as a valuable predictive tool in some cases. It is fairly simple, flexible, and not highly parameterized so that it can be applied to systems for which little information is available. For example, the reactions $\text{H} + \text{HX}$ ($\text{X} = \text{Cl}, \text{Br}, \text{I}$) yield some excited halogen atoms as product. The fraction of excited atoms produced is unknown. The statistical theory was applied to yield the following per cent excitation in thermal (300°K) reactions: Cl ($^2P_{3/2}$) 29 %; Br ($^2P_{3/2}$) 17.5 %; I ($^2P_{3/2}$) 12 %.

The statistical theory appears to be useful for those cases where alternative products are available and one is interested in determining the ratios of possible products. The information desired is highly averaged and so the results should be fairly reliable. In addition, little information about potential energy surfaces or their crossing points is needed so the computation can be carried out quickly and efficiently.

We may thus consider the statistical theory as an "interim" theory—a semi-quantitative model to be used in a comparative or predictive way until we have the ability to make accurate *a priori* calculations of all kinetic quantities of interest.

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¹ D. L. Bunker, *Theory of Elementary Gas Reaction Rates*, (International Encyclopedia of Physical Chemistry and Chemical Physics, Topic, 19, Vol. 1) (Pergamon Press, New York, 1966).

² K. J. Laidler and J. C. Polanyi, *Prog. Reaction Kinetics* 1965, 3, 1.

³ J. Ross, ed., *Adv. Chem. Physics*, vol. X, (Interscience, New York, 1966).

⁴ T. T. Warnock, R. B. Bernstein, and A. E. Grosser, *J. Chem. Physics*, 1967, 46, 1685.

⁵ S. Datz and R. E. Minturn, *J. Chem. Physics*, 1964, 41, 1153.

⁶ K. R. Wilson, G. H. Kwei, J. A. Norris, R. R. Hern, J. H. Birely and D. R. Herschbach, *J. Chem. Physics*, 1964, 41, 1154.

- ⁷ J. C. Polanyi, *Chem. in Britain*, 1966, **2**, 151.
- ⁸ R. J. Suplinskas, *Thesis*, (Brown University, 1965).
- ⁹ J. L. Magee, *J. Chem. Physics*, 1940, **8**, 687.
- ¹⁰ N. C. Blais and D. L. Bunker, *J. Chem. Physics*, 1962, **37**, 2713; *ibid.*, 1963, **39**, 315.
- ¹¹ J. C. Polanyi and S. D. Rosner, *J. Chem. Physics*, 1963, **38**, 1028.
- ¹² M. Karplus and L. M. Raff, *J. Chem. Physics*, 1964, **41**, 1267.
- ¹³ M. Karplus, R. N. Porter and R. D. Sharma, *J. Chem. Physics*, 1964, **40**, 2033.
- ¹⁴ J. C. Keck, *J. Chem. Physics*, 1958, **29**, 410.
- ¹⁵ O. B. Firsov, *Zhur. Eksperim. i. Teor. Fiz.* 1962, **42**, 1307, (Eng. trans.; *Soviet Phys.—JEPT* 1962, **15**, 906).
- ¹⁶ J. C. Light, *J. Chem. Physics*, 1964, **40**, 3221.
- ¹⁷ E. E. Nikitin, *Theor. Expt. Chem., U.S.S.R.*, 1965, **1**, 428.
- ¹⁸ F. T. Wall, L. A. Hiller Jr., and J. Mazur, *J. Chem. Physics*, 1958, **29**, 255; *ibid.*, 1961, **35**, 1284.
- ¹⁹ K. E. Shuler, T. Carrington and J. C. Light, *Appl. Optics*, 1965, supp. 2, 81.
- ²⁰ S. Glasstone, K. J. Laidler and H. Eyring, *The Theory of Rate Processes*, (McGraw-Hill, New York, 1941).
- ²¹ E. Fermi, *Progr. Theor. Physics, Japan.*, 1950, **5**, 570.
- ²² J. C. Light, *J. Chem. Physics.*, 1964, **40**, 3221.
- ²³ P. Pechukas and J. C. Light, *J. Chem. Physics*, 1965, **42**, 3281.
- ²⁴ J. C. Light and J. Lin, *J. Chem. Physics*, 1965, **43**, 3209.
- ²⁵ P. Pechukas, J. C. Light and C. Rankin, *J. Chem. Physics.*, 1966, **44**, 794.
- ²⁶ J. Lin and J. C. Light, *J. Chem. Physics*, 1966, **45**, 2545.
- ²⁷ B. C. Eu and J. Ross, *J. Chem. Physics*, 1966, **44**, 2467.
- ²⁸ P. Pechukas, *Thesis*, (University of Chicago, 1966).
- ²⁹ F. Wolf, *J. Chem. Physics.*, 1966, **44**, 1619.
- ³⁰ K. Yang, J. D. Pader and G. L. Hassell, private communication.
- ³¹ J. C. Polanyi, *J. Quant. Spectr. Radiative Transfer* 1963, **3**, 471.
- ³² J. R. Airey, R. R. Getty, J. C. Polanyi and D. R. Snelling, *J. Chem. Physics*, 1964, **41**, 3255.
- ³³ J. R. Airey, F. D. Findley and J. C. Polanyi, *Can. J. Chem.* 1964, **42**, 2193.
- ³⁴ The recent (unpublished) work of J. C. Polanyi *et al.*, provides some information on the limits of the initial distribution.
- ³⁵ D. Beck, E. F. Gruen and J. Ross, *J. Chem. Physics*, 1962, **37**, 2895.
- ³⁶ A. E. Grosser, A. R. Blythe and R. B. Bernstein, *J. Chem. Physics*, 1965, **42**, 1268; A. E. Grosser and R. B. Bernstein, *ibid.*, 1965, **43**, 1140.
- ³⁷ S. Datz, D. R. Herschbach and E. H. Taylor, *J. Chem. Physics*, 1961, **35**, 1549; D. R. Herschbach, *Disc. Faraday Soc.*, 1962, **33**, 149.
- ³⁸ R. J. Suplinskas, private communication.
- ³⁹ O. Karl, P. Kruus and J. C. Polanyi, *J. Chem. Physics*, 1967, **46**, 224.
- ⁴⁰ General reviews of ion-molecule reactions are available in *Adv. Chem.*, vol. 58, (Amer. Chem. Soc. Publ.) 1966, Washington, D.C.
- ⁴¹ C. F. Giese and W. B. Maier II, *J. Chem. Physics*, 1963, **39**, 739.
- ⁴² M. von Koch and L. Friedman, *J. Chem. Physics*, 1963, **38**, 1115.
- ⁴³ C. F. Giese, *Adv. Chem. Physics.*, (Interscience, New York, 1966).
- ⁴⁴ P. Langevin, *Ann. Chem. Physics.*, 1905, **5**, 245; G. Gioumousis and D. P. Stevenson, *J. Chem. Physics*, 1958, **29**, 294.
- ⁴⁵ M. E. Wacks, *J. Res., Nat. Bur. Stand.* 1964, **68A**, 631.
- ⁴⁶ L. Friedman, *Adv. Chem.*, vol. 58, (Amer. Chem. Soc., Washington, D.C. 1966), p. 87.
- ⁴⁷ J. H. Futrell and F. P. Abramson, *Adv. Chem.*, vol. 58, (Amer. Chem. Soc., Washington, D.C. 1966), p. 107.
- ⁴⁸ There are exceptions such as $O^+ + N_2$ (cf. C. Giese, ref. (40), p. 20) for which an activation energy appears to occur for an exothermic reaction. In these cases the normal definition of the ion-molecule complex appears to be in error.